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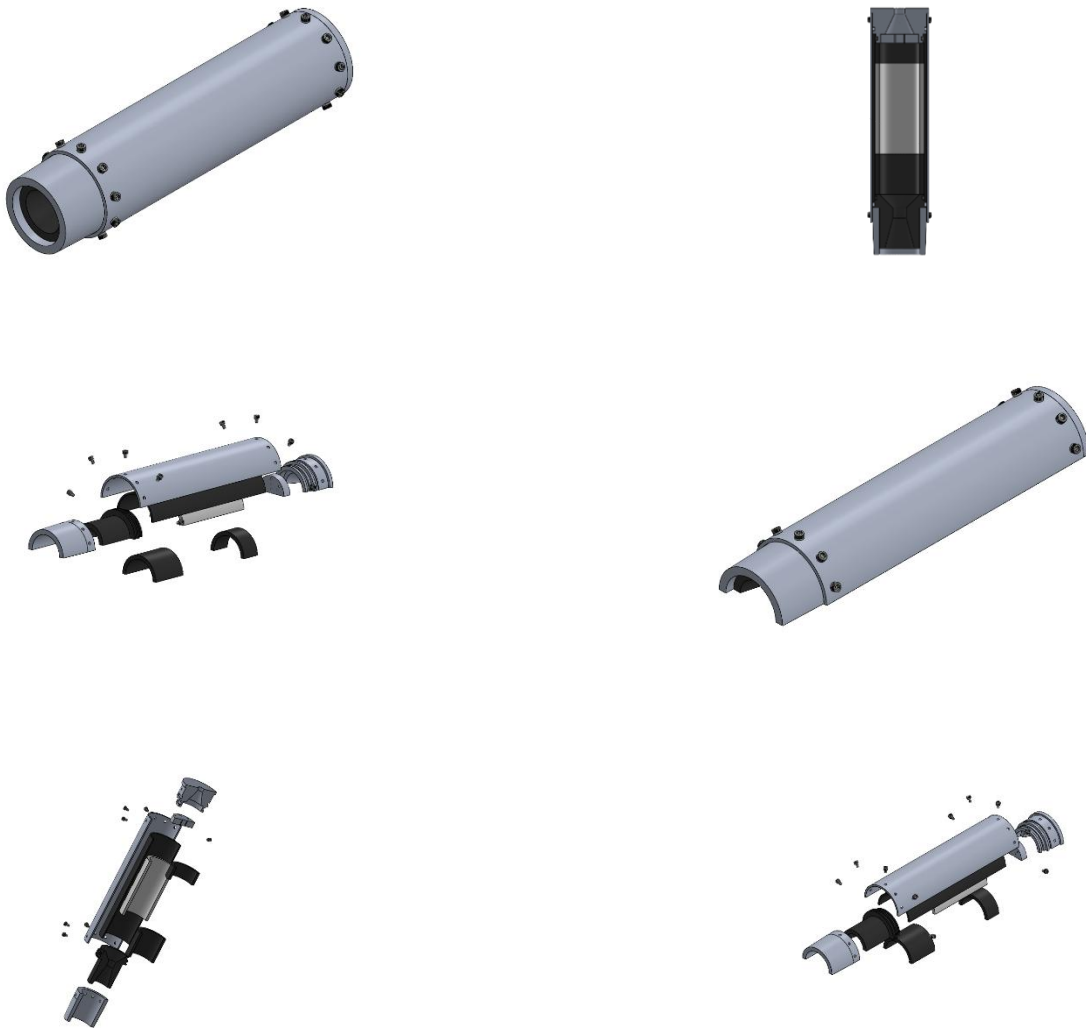
Github: <https://github.com/nrobertson1453/portfolio>

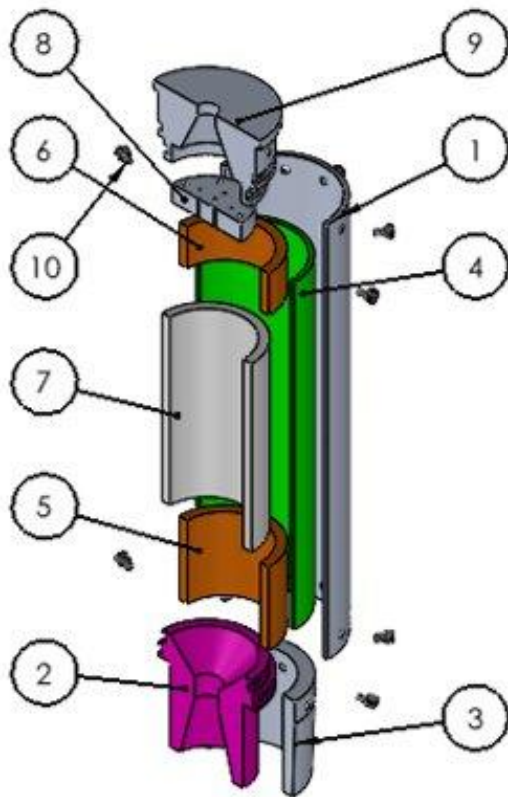
Combustion Chamber, Hybrid Rocket Motor N-class and K-class

Fall 2024-Fall 2025

- Contributed to the design and development of the combustion chamber of an N-class hybrid rocket motor (20,000 N-s of total impulse) Fall 2024 and Spring 2024
- Spring 2025 spearheaded the design of a K-class motor (2,500 N-s of total impulse). Made extensive and iterative design decisions regarding safety, thermal and structural analysis, optimization, material selection, fabrication, all within tight time and cost constraints, and in close coordination with several other design teams. Produced a complete CAD assembly for the K-class motor.

K-class Combustion Chamber CAD Assembly (files in GitHub):





BOM Table		
ITEM NO.	PART NUMBER	QTY.
1	Casing	1
2	Nozzle	1
3	Nozzle Casing	1
4	Fuel Grain Liner	1
5	Post Combustion Chamber Liner	1
6	Pre Combustion Chamber Liner	1
7	Fuel Grain	1
8	Injector Plate	1
9	Injector	1
10	Bolts	20

PEP Racer: Potential Energy Powered Racer

- Designed, tested, and manufactured a “mouse-trap” style car utilizing propellers, springs, and a ramp with strict mass, part, and size constraints with team of 6. Singlehandedly redesigned car to function during iteration. Team scored one of the highest speeds in class of 168.



Fall 2024

- ## Project Poster:



Evan Daniel, Abel Saberhagen, Omer Mirza, Nolan Robertson, Jon Buszard, Austin Phan

FUNCTION

FINDINGS AND FUTURE

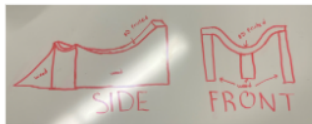
Our project was ultimately successful in addressing the issue of inaccessible bike parking by developing a functional bike parking notification system. If we had more time or better materials, we would use a waterjet to produce more precise parts, incorporate additional 3D printing for better cohesion within the system, and fully integrate the system with the website wirelessly to enhance real-time functionality.

1. Our design must be user-friendly and be effective for those traveling to classes.
2. Bicycle users must be able to easily see the availability of a parking space before they get to class.
3. The design must be compact enough to fit within standard bike parking areas
4. The sensor and notification system must work consistently
5. The system should significantly improve bike parking organization and increase convenience.

Wu, P. (n.d.). Undergraduate Commuter Student Challenges and Struggles. ERHC. Retrieved November 4, 2024, from <https://eric.ed.gov/?id=ED512476>

AASHE. (n.d.). University of Virginia OP-18: Commuter Model Split: The Sustainability Tracking, Assessment & Rating System. Retrieved November 4, 2024, from <https://reports.aashe.org/systems/university-of-virginia-report/2021-03-04/C%20transportation/C%2018>

Slite Theft Prevention - Public Safety. (n.d.). Macalester College. Retrieved November 4, 2024, from https://www.macalester.edu/safety/safety-prevention/bike_theft_prevention/



The image consists of two side-by-side photographs of a curved wooden structure, possibly a model of a ship's hull or a water channel. The structure is made of light-colored wood and has a blue interior lining. It is shown from two different angles, highlighting its curved shape and the blue lining. The structure is placed on a wooden surface.

Bridge

Fall 2025

- Designed and manufactured a “bridge” able to hold 50 lbf for a class competition utilizing 2 in³ of 3D ABS plastic with a team of 6. Project involved the use of progressive design, CAD, material and structural analysis, and team coordination.

Our final design and SolidWorks FEA analysis

